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Evolution of Diagnostic Neuroradiology from 1904 to 1999¹

Neuroradiology began in the early 1900s soon after Roentgen discovered x rays, with the use of skull radiographs to evaluate brain tumors. This was followed by the development of ventriculography in 1918, pneumoencephalography in 1919, and arteriography in 1927. In the beginning, air studies were the primary modality, but this technique was supplanted by angiography in the 1950s and 1960s. The first full-time neuroradiologist in the United States was Cornelius G. Dyke at the New York Neurological Institute in 1930. Neuroradiology took a firm hold as a specialty in the early 1960s when Dr Juan M. Taveras brought together fourteen neuroradiologists from the United States and Canada to establish the nucleus of what was to become the American Society of Neuroradiology, or ASNR. This society's initial goals were to perform research and to advance knowledge within the specialty. Neuroradiologists initially were able to diagnose vascular disease, infections, tumors, trauma, and alterations in cerebrospinal fluid flow, because the brain structure was invisible. Neuroradiology was forever changed with computed tomography (CT) because the brain structure became visible. Soon thereafter, magnetic resonance (MR) imaging was developed, and it not only provided anatomic but also made possible vascular and physiologic functional imaging.

Imaging of the central nervous system began with radiographs (plain roentgenograms) of the skull, following the discovery of x rays by Roentgen in 1895. In the early 1900s, Dr George Pfahler (1) in the United States (Fig 1) and Dr Arthur Schüller (2) in Europe used emulsion-coated glass plates to obtain images to diagnose cerebral tumors and other lesions. The first major innovation occurred in 1918 with the introduction of ventriculography by Dr Walter Dandy (3), a prominent neurosurgeon at Johns Hopkins Hospital. Following this, in 1919 he reported on the first use of pneumoencephalography (4).

In 1927, Egas Moniz, a Portuguese neurologist, introduced opacification of the carotid artery with contrast medium by using a solution of sodium iodide. This technique became fully developed in the 1930s with the advent of direct percutaneous puncture and iodinated organic contrast medium (Fig 2) (5).

The first full-time neuroradiologist in the United States was Dr Cornelius G. Dyke, who assumed this position at the Neurological Institute of Columbia Presbyterian Medical Center in New York in 1930. He remained in this position until his untimely death of leukemia in 1943, but during that time he published numerous pioneering articles and a classic monograph in collaboration with Dr Leo M. Davidoff, the prominent neurosurgeon, in 1937—*The Normal Encephalogram* (6).

In 1952, Dr Juan M. Taveras (Fig 3) became head of neuroradiology at the Neurological Institute of the Columbia Presbyterian Medical Center in New York, and from these beginnings, the envelope of neuroradiology was expanded and the future cadre of neuroradiologists had its beginning. Dr Taveras became the father of neuroradiology by leading the charge for recognition of neuroradiology as a specialty; he was primarily responsible for the formation of the American Society of Neuroradiology. Dr Taveras recognized the need for an organized group to represent American neuroradiology to promote the advance of research and knowledge in this field. At a historic dinner meeting at Keen's Chop House in New York City held on April 19, 1962, he brought together a group of 14 neuroradiologists from the United States and Canada (Table). It was this invited group

who, by responding positively to Dr Taveras' recommendation, became founding members of the American Society of Neuroradiology.

Juan Taveras together with M. M. Schechter was further responsible for the recognition of the importance of our specialty by both neurologists and neurosurgeons in the United States through their many scientific presentations and service on joint committees with their clinical colleagues. In addition, Dr Taveras developed the first National Institutes of Health-sponsored fellowship training program in neuroradiology at the Neurological Institute in 1960. Soon thereafter, Dr Schechter developed the second fellowship program at the Albert Einstein College of Medicine in New York City. These two remarkably able and energetic individuals were the two major forces in the development and recognition of the specialty.

An advantage of choosing neuroradiology to practice at that time was that it seemed almost like general practice of radiology, since the daily schedule included interpreting conventional radiographs of the skull, sinuses, and spine, as well as performing interventional procedures such as myelography, encephalography, and cerebral angiography. Skull radiographs in 1960 were read to identify pineal shift, sellar enlargement due to tumor, or erosion from elevated intracranial pressure, as well as to detect abnormal calcifications, bone erosion, or bone destruction; metastatic lesions; and hyperostosis of the calvaria. Spine radiographs were evaluated for degenerative vertebral changes, disk space narrowing, erosion of the vertebral end plates, bone destruction or production, a fracture, loss of a pedicle, interpediculate widening, vertebral body scalloping, and scoliosis or kyphosis.

Myelography was performed with iophendylate (Pantopaque), an oily radiopaque contrast medium that had to be removed from the subarachnoid space following the examination, since residual Pantopaque could lead to the development of arachnoiditis. During myelography, the patient's head needed to be fully extended, not flexed, when the patient was tilted head downward to examine the thoracic or cervical region, to prevent the oily contrast medium from spilling into the cranial cavity.

Myelography was used to evaluate disk disease (7) and extradural, intradural extramedullary, and/or intramedullary lesions or to identify a block to cerebrospinal fluid flow by an intraspinal lesion (8).

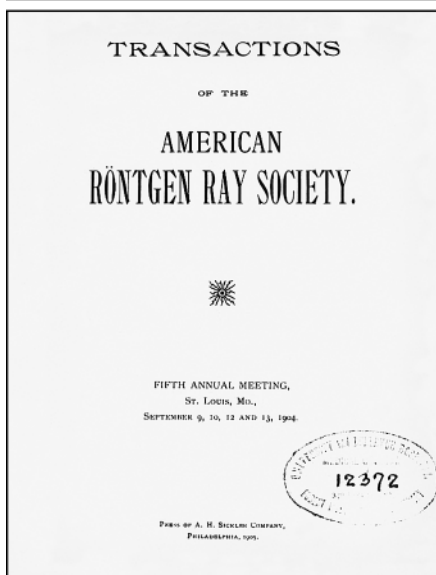
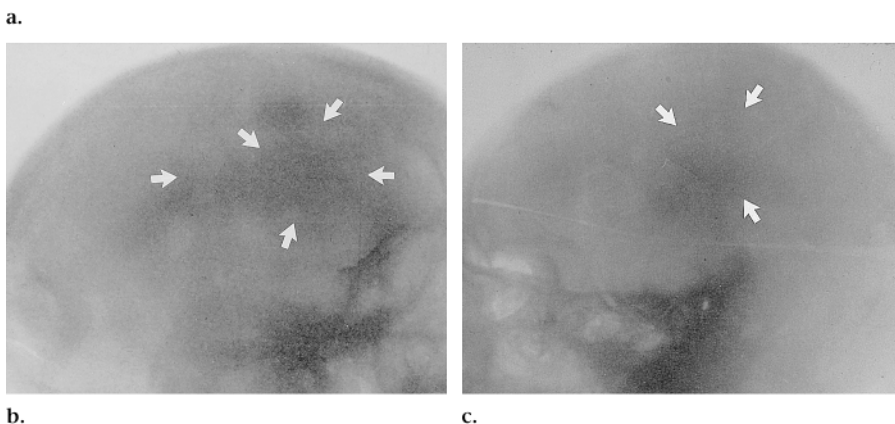


Figure 1. (a) Photograph shows the first publication of skull radiographs in diagnosing tumor or infarct in *Transactions of the American Röntgen Ray Society* in 1904. (b, c) Lateral skull radiographs reveal opacity, outlined by arrows, considered to be tumor, which is probably hair with grease or tangles, as considered by H. B. Baker, Jr, a prominent neuroradiologist of the Mayo Clinic. (Reprinted, with permission, from the American Roentgen Ray Society.)



Founding Members of the American Society of Neuroradiology

Norman E. Chase, MD
William N. Hanafey, MD
Colin B. Holman, MD
Eugene V. Leslie, MD
Thomas H. Newton, MD
D. Gordon Potts, MD
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Giovanni Di Chiro, MD*
Fred J. Hodges III, MD
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Donald L. McRae, MD*
Harold O. Peterson, MD*
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In this procedure, fluoroscopic and radiographic imaging were performed with the patient lying prone on a 90°-to-90° tilting table. Images were obtained in the posteroanterior, oblique, and cross-table lateral projections for accurate localization. In the upper thoracic or lower cervical regions, a spurious or transient block might be encountered when the patient was tilted head downward because of the presence of hypertrophic posterior vertebral marginal spurs or infolded ligamenta flava, which, with the

neck hyperextended, would compromise an already tight thecal sac. If the apparent block still persisted, when the patient was returned to the horizontal position, turning the patient's spine obliquely sometimes succeeded in relieving the obstruction to the flow of the bolus of contrast medium.

Pneumoencephalography was performed by successively injecting small volumes of air via lumbar puncture and then removing small volumes of cerebrospinal fluid with the patient sitting up-

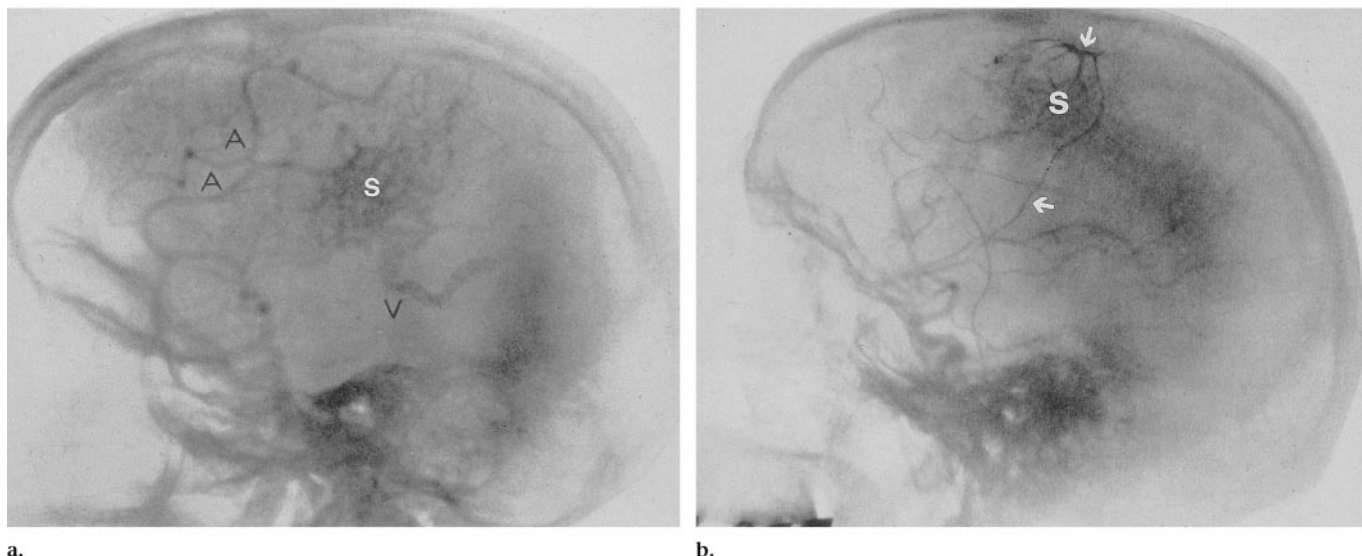


Figure 2. Lateral common carotid arteriograms obtained during the early phase. (a) Image reveals anterior cerebral arteries (A) with an early vascular stain (S) and an early draining vein (V), which indicates glioblastoma multiforme. (b) Image demonstrates a prominent middle meningeal artery (arrows) inducing an early vascular stain (S) in a meningioma. (Reprinted from reference 5.)

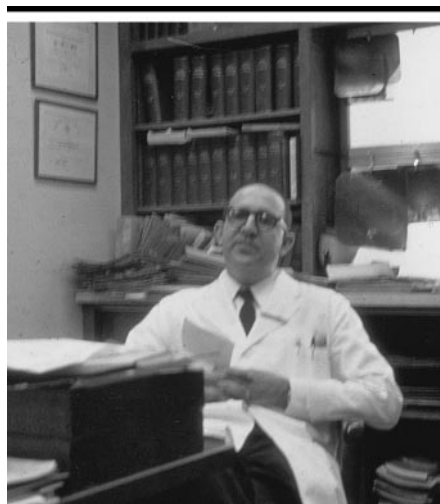


Figure 3. Photograph shows Dr Juan Taveras, then head of Neuroradiology at the Neurological Institute of the Columbia Presbyterian Hospital, in his office at the Neurological Institute in New York in 1962. (Image courtesy of Juan M. Taveras, MD, Department of Radiology, Massachusetts General Hospital, Boston.)

right and the head flexed (Fig 4). Pneumoencephalography was used primarily to determine the presence and extent of posterior fossa or cerebellopontine angle tumors, pituitary tumors, and intraventricular masses (9). It was also used to rule out the presence of lesions affecting the cerebrospinal fluid spaces in patients with possible communicating hydrocephalus or dementia (Fig 5) (10). After the injection

of a sufficient quantity of air, the patient was rotated, somersaulted, or placed in a decubitus position to depict the entire ventricular system and subarachnoid spaces. These patients were often uncomfortable, developed severe headaches, and became nauseated or vomited.

At the Neurological Institute, a somersaulting chair was developed by Dr Gordon Potts in 1965 (11), so that manual rotation was no longer required and better filling of ventricles could be attained. Other somersaulting apparatuses for pneumoencephalography were developed by Dr Kurt Amplatz at the University of Minnesota and by Elema-Schonander (the Mimer chair) in Sweden. Use of this uncomfortable procedure was abruptly terminated in the early 1970s when CT became available, because CT could depict the ventricular system and intracranial subarachnoid spaces directly without the need to introduce air into the subarachnoid spaces. Pneumoencephalography was rarely used in the assessment of supratentorial tumors.

If an obstruction of the ventricular system was known or suspected or if the intracranial pressure was elevated, ventriculography was used for localizing and defining the extent of supratentorial, third ventricular, or posterior fossa lesions (12). For this technique, a catheter was inserted into the lateral ventricles via a burr hole or through a patent anterior fontanelle, and air was injected. Ventriculography gradually disappeared from use as interpretation of angiograms im-

proved, and particularly when vertebral angiography became easier and safer to perform by using the catheter technique via the femoral artery.

Cerebral angiography in the early 1960s was typically performed by means of direct puncture of the common carotid artery, the vertebral artery, or the brachial artery (with retrograde injection) (13,14). In the late 1960s and early 1970s, cerebral angiography was performed predominantly by means of selective catheter angiography following puncture of the femoral artery. Cerebral angiography was first performed by means of a direct stick into the common carotid artery. Once the puncture was made, the needle was advanced with the use of direct visualization, with the needle antegradely within the arterial lumen to make its position more stable. A flexible plastic tube was connected to the needle threaded with a two-way stopcock. This took pressure off the needle so it was stabilized within the arterial lumen.

The common carotid artery in the neck was first imaged by means of manual injection of a small aliquot of iodinated ionic contrast medium to be certain the needle was well positioned in the arterial lumen and to avoid a subintimal or wall injection. Serial angiography was then performed to evaluate intracranial vessels.

Images were obtained on a manual-pull or a power-driven (Sanchez Perez) film changer. Three or four film hard copies were obtained, and separate injections

were required for each plane. In the 1960s, a serial roll film changer was used at the Neurological Institute to enable the acquisition of as many images as required, usually two per second for 3 seconds and then one per second for 6 seconds.

The lesions we looked for included vascular lesions, such as aneurysms, arteriovenous malformations, arterial or venous occlusions, and vasculopathies (abnormalities of the blood vessel wall); cerebral tumors; traumatic lesions such as subdural or epidural hematomas; cerebral contusions; intracerebral hematomas; and traumatic fistulas or aneurysms. A lesion was revealed by the midline shift of arteries or veins and by the localized displacement of arteries or veins. The sylvian triangle, demarcated on the lateral projection image by the middle cerebral artery and its branches, aided in tumor localization. Midline shift of the anterior cerebral artery occurred with frontal, parietal, or temporal masses, whereas deep lesions involving the thalamus or medial temporal region resulted in shift of the internal cerebral vein. Shifts of the anterior cerebral artery were characterized as rounded (anterior frontal mass), angled (temporal or deep frontal mass), or square (posterior frontal or parietal lesion) (15).

The various gliomas could be identified by relating vascular patterns to circulatory changes in the different phases (arterial, intermediate, or venous) of the angiogram. In glioblastoma multiforme, for example, a tumor circulation with sinusoidal channels and a typical contrast enhancement pattern appeared early in the arterial phase and disappeared before the normal veins appeared (Figs 2a, 6) (16). Early opacification of venous drainage from the tumor was a frequent finding (Figs 2a, 6). Most other gliomas of a lesser grade manifested as a lesion with little or no abnormal vascularity.

Meningiomas produced a characteristic well-defined blush, with blood supply coming from the internal and external carotid arteries (Fig 2b). The diffuse enhancement pattern was sharply demarcated, appeared early, and remained late until after the veins had emptied. Metastatic lesions could be very vascular or hypovascular. Vascular lesions appeared as an intense blush and/or collection of abnormal vessels that developed in the late arterial, intermediate, or early or late venous phase.

The abnormal vascular patterns were categorized and used to distinguish the various types of hypervascular lesions. In addition to vascular displacements, vari-

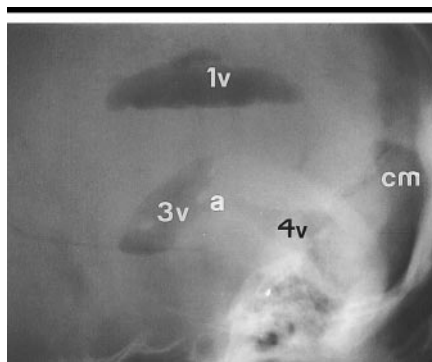


Figure 4. Normal lateral pneumoencephalogram. The patient is upright in a chair, and the first 12 mL of air is injected after a removal of 6 mL of fluid. Structures identified include the cisterna magna (cm), fourth ventricle (4v), sylvian aqueduct (a), third ventricle (3v), and lateral ventricle (lv).

ations in vascular dynamics also proved useful. These included delayed filling of veins, as well as early filling of veins and delayed filling of arteries (17). All these changes in vascular dynamics were used as the basis for today's perfusion imaging.

Circulation time was measured according to the method of Torgny Greitz, MD, in Stockholm (18) and was influenced by vascular lesions, tumors, aging, and increased intracranial pressure. The normal circulation time, as measured by Greitz (18), was approximately 4.13 seconds. Subsequently, Leeds and Taveras (17,19) not only measured circulation time to be 4.3 seconds by using the method of Greitz (18), they also measured local circulatory dynamics in arterial, intermediate (capillary), and venous phases and identified the variable influences of aging on local circulatory dynamics, from the pediatric to the elderly patient.

A vascular blush or enhancement pattern might also be observed in the patient with ischemic changes (Fig 7). This is the result of luxury perfusion and changes in cerebral autoregulation (20). To elaborate further, as a result of loss of cerebral autoregulation, the capillary bed is opened, and the dilution that normally affects the hypertonic contrast medium is lost. The contrast medium within the capillary bed and draining vein is therefore as bright as it is during the arterial phase (Fig 7b). In the patient with normal cerebral autoregulation, the hypertonic contrast medium within the capillaries becomes diluted, such that the intensity of the contrast medium is reduced in the veins as compared with that in the arteries (20).



Figure 5. Lateral pneumoencephalogram shows the optic recess (O) in the third ventricle in a patient with headache and evidence of an incisural block. Patient is supine with head up. Note the fluid levels in the enlarged lateral ventricles. No air is observed beyond the prepontine cistern (arrow). (Image courtesy of Irvin I. Kricheff, MD, Department of Radiology, New York University Medical Center.)

Direct puncture of the vertebral artery, a technique pioneered by Phillip Sheldon (21) in Oxford, England, could be performed but was difficult to do, since the needle could not be threaded into the artery. The vertebral artery and aortic arch could also be examined by means of puncturing the right brachial artery and threading the needle retrogradely up the artery (13,14). This became feasible once power injectors were developed that could deliver the bolus of contrast medium retrogradely into the subclavian artery and brachiocephalic trunk. The right brachial arterial injection could be used to examine the aortic arch on the right side, as well as the brachiocephalic trunk, and to depict the right common carotid and right vertebral arteries. To examine the left side of the aortic arch and left vertebral artery, the left brachial artery received the injection of contrast medium by means of a power injector.

In the 1960s, transfemoral catheter placement into each branch off the aorta was introduced and replaced direct sticks of the carotid and brachial arteries (22). This made it possible to selectively examine the branches of the aorta to the brain. The information accumulated about the vascular supply of the posterior fossa, and in particular of the veins of this structure (Yun Peng Huang and Bernard Wolf of Mount Sinai Hospital in New York [23,24]), paved the way to a better understanding of the posterior fossa.

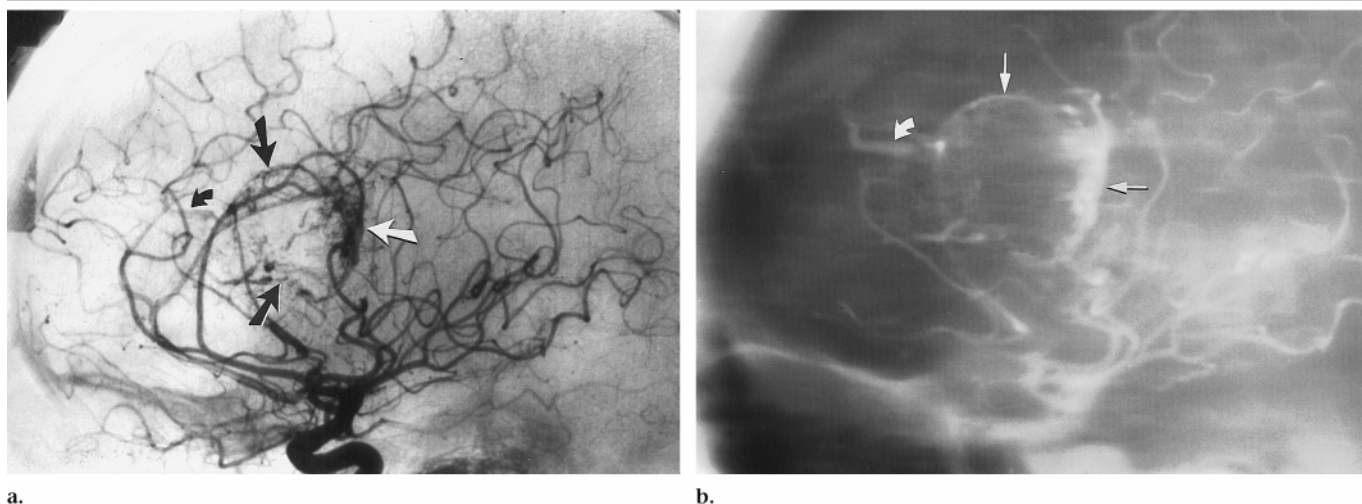


Figure 6. Cerebral angiograms show glioblastoma multiforme. (a) Lateral subtracted image obtained during the midarterial phase reveals early vascular stain (straight arrows) and the presence of a barely perceptible draining vein (curved arrow), which is better appreciated in (b) lateral angiotomogram, which reveals the abnormal vascular stain to advantage (straight arrows) and highlights the early draining vein (curved arrow).

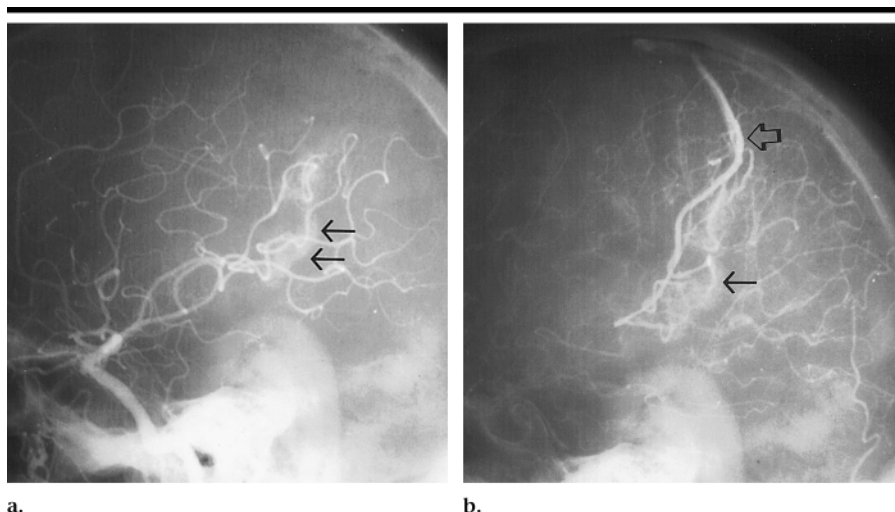


Figure 7. Lateral carotid arteriograms show vascular blush in a patient with an early middle cerebral infarct. (a) Image obtained during the midarterial phase shows a vascular blush (arrows) around the distal ascending branch of the sylvian triangle. (b) Image obtained during the intermediate phase shows an early draining vein (open arrow) that has emerged from the blush (solid arrow). Note the intensity of the contrast medium in the vein because of lack of dilution.

Knowledge of the normal anatomy of the veins in the posterior fossa and of the pathologic alteration of these vessels caused by lesions opened the door to much improved localization and diagnosis of lesions in the posterior fossa and cerebellopontine angle (Fig 8). This led to a reduction in the need for pneumoencephalography and ventriculography and enabled more accurate surgical planning, which in turn translated into reduced intraoperative and postoperative morbidity and mortality. Other technologic advances to improve cerebral angiography included subtraction (Fig 6a) and magnification

(25) angiography and angiotomography (Fig 6b) (26).

Despite the considerable advances made in neuroradiology as we approached the end of the momentous decade of the 1960s, there were still many lesions of the central nervous system we were unable to evaluate by using these techniques. These included multiple sclerosis and other white matter lesions, as well as other degenerative and developmental lesions. Inflammatory lesions such as encephalitis and early cerebral abscesses were also difficult to evaluate. In addition, pituitary lesions were not seen with the

frequency with which they were subsequently diagnosed by means of CT (27) and then even more so with MR imaging (28). From the 1940s through the development of CT in 1971, there were only incremental improvements in neuro-radiologic methods, techniques, and knowledge. In contrast, in the period since 1971, much more substantial changes have occurred more rapidly.

CT, which was initially called "computerized axial tomography" or "CAT scanning," was developed by Godfrey Hounsfield at the EMI Laboratories in England (29) and was introduced clinically by Jamie Ambrose of the Maida Vale Hospital in London in 1971 (30). Suddenly, the way the brain could be examined was instantaneously and forever changed. Cerebral angiography and pneumoencephalography were invasive procedures and posed risks to patients, so these procedures were performed only when necessary and were not repeated with any frequency, except periodically to follow the course of vascular lesions such as aneurysms and arteriovenous malformations.

With the advent of CT, the internal structure of the brain could be depicted directly, and a new era in cerebral studies dawned. It was then possible to understand and diagnose almost any type of cerebral lesion noninvasively (Fig 9). It was also then possible to diagnose intrasellar, parasellar, and suprasellar lesions, and because of this the number of such cases diagnosed increased substantially (31). The paranasal sinuses and orbits could be examined much more accu-

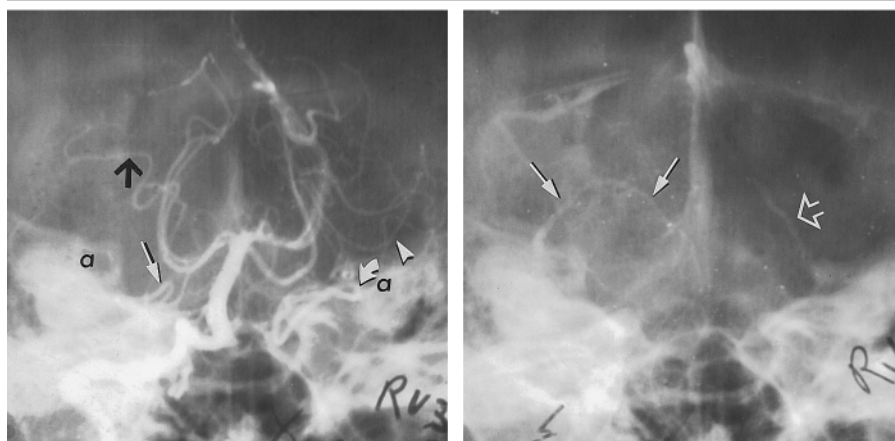


Figure 8. Right vertebral arteriograms demonstrate tumor in the right cerebellopontine angle. (a) Anteroposterior image obtained during the arterial phase reveals elevation of the marginal artery (black arrow) of the superior cerebellar artery compared with the normal configuration on the left (arrowhead). The anterior inferior cerebellar artery (straight white arrow) on the right is foreshortened and displaced inferiorly from the internal auditory canal (a), whereas on the left the artery has a normal configuration with a loop (curved white arrow) seen at the internal auditory canal. (b) Anteroposterior image obtained during the venous phase shows displaced petrosal vein (solid arrows) arched over the tumor on the right, whereas the left petrosal vein (open arrow) is normal.

rately (32). Suddenly, we had to learn the internal structure of the orbit.

The ventricles, interfaces between gray and white matter, and calvaria could be depicted directly in considerable detail by means of CT. Furthermore, the addition of intravenous contrast medium resulted in the depiction of arteries and veins, dura mater, meninges, and abnormal vascular patterns of tumors. The evaluation of acute cerebral changes resulting from trauma, infection, hemorrhage, or metabolic alteration was possible, and this in turn meant that therapy could be started immediately in many patients, with a resultant decrease in morbidity and increase in improved outcomes.

For example, the early diagnosis of subdural empyema could be made, whereas before the advent of CT this condition was fatal in more than 50% of patients (33). In the CT era, only one of 32 patients with subdural empyema in one reported series (34) of patients died, and this patient had venous thrombophlebitis. Intracranial hemorrhage could be recognized, diagnosed with certainty, and localized (35). Angiography could then be properly directed in preparation for surgery. Cerebral abscess could be recognized early in its development and treated in a timely and effective manner (Fig 10) (36). The presence of hydrocephalus or tumor was appreciated, and treatment could be planned accurately and safely.

Thus, CT marked the beginning of a new era in which diagnosis could be made (Fig 11) and therapy thereby instituted earlier, which reduced morbidity and mortality and improved outcomes. A further important advantage of CT over earlier imaging technology was that it could be repeated as often as needed to follow the course of structural lesions.

The introduction of CT put an end to the need for air studies, such as pneumoencephalography and ventriculography. In addition, besides enabling examination of the brain, CT could be used to look for osseous lesions of the paranasal sinuses and orbits, as well as of the spine. Intravenously administered contrast medium could be used to enhance the dura mater, which could aid in the evaluation of cervical or lumbar disk herniation (37).

In the 1970s, metrizamide (Amipaque), the first nonionic isosmolar contrast medium containing iodine, was introduced for myelography (38). It made examination easier because it did not need to be removed, as it was absorbed into the bloodstream over time. This nonionic contrast medium also did not cause arachnoiditis and could be used in combination with CT to examine intraspinal lesions (CT myelography) (39). It could further be used with CT of the brain for opacification of the intracranial cisterns (CT cisternography) to evaluate obstruc-



Figure 9. Transverse CT scan obtained after the administration of contrast material depicts a large ring lesion (arrow) in the left frontal corticomedullary junction. The patient had developed right hemiparesis 10 months after radiation therapy for a right parasagittal lesion. No ring was seen on the CT scan (not shown) obtained before the administration of contrast material, which excludes abscess, metastasis, or glioblastoma. This was supported by lack of substantial edema and the thickening of the medial margin of the ring lesion. At surgery, radiation necrosis was confirmed.



Figure 10. Transverse postcontrast CT scan obtained in the inferior frontal region shows cerebral abscess after fracture of frontal bone. A ring lesion (curved arrow), which was also seen on the precontrast CT scan (not shown), is demonstrated. Although no thinning is seen on the medial margin, the budding of a new lesion from the anterior margin of the ring (straight arrow) suggests an abscess, which was confirmed at surgery.

tive lesions (40). It could also be used to evaluate cerebrospinal fluid leaks.

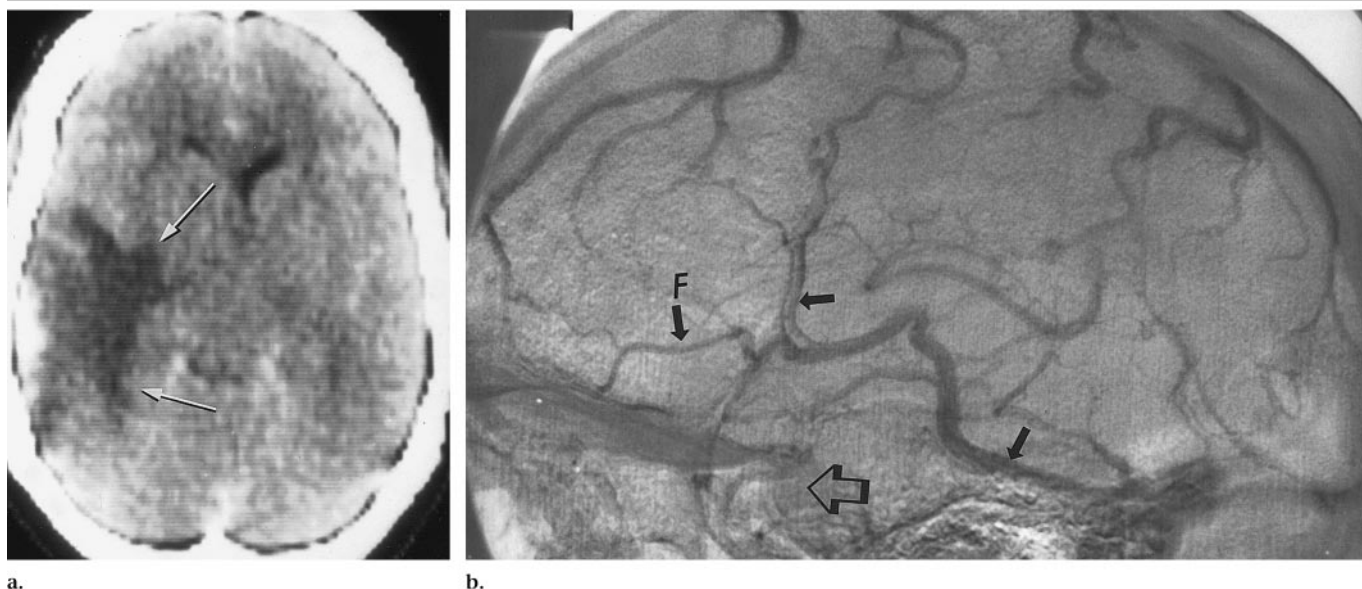


Figure 11. Images obtained in a young adult woman with lupus erythematosus who presented with left hemiparesis. **(a)** Transverse contrast medium-enhanced CT scan reveals a large hypoattenuation zone (arrows) affecting temporal white matter without any evidence of contrast enhancement, which suggests a venous infarct. **(b)** Lateral right common carotid arteriogram obtained during the venous phase with subtraction reveals occlusion of the sylvian vein by its absence (open arrow), and major venous drainage is via the vein of Labbe (solid arrows) and to a lesser extent a frontal vein (*F*). A right common carotid arteriogram (not shown) obtained during the arterial phase revealed no abnormality.

Progressive improvements over time, including improved signal-to-noise ratio and contrast-to-noise ratio with markedly reduced data acquisition and processing times, have resulted in substantial improvements in imaging quality and applications. New scanners have been developed with spiral motion and multisection capabilities that allow faster acquisition times with ever thinner sections that yield detailed anatomic and vascular studies (41). Aneurysms can be depicted directly and more completely (42). Perfusion imaging techniques are currently advancing and rival those available with MR imaging (43). Thus, CT can image the brain, orbits, and spine, and our knowledge of congenital, developmental, white matter, degenerative, traumatic, inflammatory, and neoplastic lesions has progressed by a quantum leap.

CT examination of the neck has allowed evaluation of the deep and superficial cervical lymph nodes. The aerodigestive tract can be examined, and the congenital and developmental lesions of the soft tissues of the neck can be diagnosed and localized (44–46). Evaluation of the cervical lymph nodes made possible by CT has led to more accurate classification of lymph node groups and has improved tumor grading and staging of head and neck neoplasms (47–49). CT has also led to similar improvements in

the diagnostic accuracy and staging of tumors of the orbits, paranasal sinuses, temporal bones, and skull base, which virtually eliminates the indications for conventional radiographs and tomograms of these regions (50).

Progress in neuroradiology continued rapidly and inexorably with the development of MR imaging in the late 1970s and early 1980s, close on the heels of CT. Particular advantages of MR imaging are that it can be performed in any plane and with various pulse sequences, which reveal more detailed cerebral anatomy, such as the red nuclei and the substantia nigra (51), than was previously possible with any earlier diagnostic imaging modality. The anatomy observed by means of MR imaging is exquisite in its contrast resolution and rivals the appearance of actual gross anatomic cerebral slices.

MR imaging has replaced CT in the imaging of the brain and spine, except in certain specific instances (52). These include acute subarachnoid hemorrhage, three-dimensional reconstructions of aneurysms with helical images, and imaging in acutely ill or agitated patients because of rapid imaging time. Patients in whom MR imaging is contraindicated are patients with a pacemaker or metallic devices, such as aneurysm clips, that may move during examination and patients with calcified or ossified intracranial lesions, including skull base tumors and

temporal bone lesions. MR imaging can be used to help obtain more specific histologic diagnoses, assess for complications (eg, hemorrhage or hydrocephalus) in the immediate postoperative interval, and depict metallic fragments within the orbit (52). The development of paramagnetic contrast media that are safe and well tolerated, coupled with the detailed anatomic depiction possible with MR imaging, has provided a unique opportunity to examine the brain in detail to explain almost any lesion that affects the human brain.

As the American Society of Neuroradiology matured, subspecialties within the society slowly developed and evolved. These included interventional neuroradiology and head and neck radiology, and then pediatric neuroradiology and the spine society. The American Society of Neuroradiology now consists of general neuroradiology and the subspecialties; these each have their own leadership but are integrated with each other, which has enabled greater growth of the specialty of neuroradiology.

Further advances that add to our knowledge and expand the horizons and applications of neuroimaging continue to be made at a rapid pace. For example, faster acquisition times have made possible new sequences such as fluid-attenuated inversion recovery, or FLAIR. This has permitted improved depiction of intra-

cranial lesions, as well as evaluation of subarachnoid fluid, so that leptomeningeal disease, subarachnoid blood, pus, or any fluid with an elevated protein content can be made visible by altering the signal intensity of the subarachnoid spaces (53). In addition, MR angiography now allows direct depiction of blood flow in arteries or veins with or without the use of contrast medium (54).

Faster imagers with stronger gradients have led to the development of echoplanar MR imaging, which is capable of *in vivo* functional neuroimaging (55). In particular, one can now observe and localize stimulated cerebral activity (eg, motor, sensory, expressive, or receptive speech and memory), which has vastly expanded our understanding of how the brain works in health and in disease and has already assumed an important place in patient examination and surgical planning (56,57).

Another technique, diffusion-weighted MR imaging, can be used to examine alterations in micromolecular water motion, which allows the early detection of conditions restricting motion of water at the cellular level (58). To date, this modality has been most effective in identifying an early stroke, thereby enabling and improving appropriate therapy planning. Diffusion-weighted imaging is now being evaluated for possible use in identifying tumors and distinguishing tumor from necrosis and edema on the basis of variations in diffusion rates and particular changes in the apparent diffusion coefficient (59). It is also being used to depict white matter tracts (Fig 12) as an aid to stimulated cerebral activity and to examine connectivity within the brain (60).

Perfusion imaging, another technique, is used to examine cerebral blood volume, cerebral blood flow, and the mean transit time of blood within the brain by means of measuring the first pass of a bolus of paramagnetic contrast medium through the brain, with multiple sections obtained in microseconds. This technique may be used to identify the more active components of tumors by revealing foci with increased uptake of contrast medium. It is hypothesized that tumor may also thus be separated from necrosis in a patient after treatment (61). In a patient who has had a stroke, a mismatch between diffusion and perfusion may be used to identify the ischemic but still viable penumbra, and this information could be important in therapeutic decision making (62).

In the examination of the spine, MR imaging has today almost completely re-

placed myelography. It is effective and accurate in the assessment of spondylosis; degenerative disk disease; traumatic lesions of the spine, including epidural hematoma; inflammatory lesions of the disk, bone, and cord; and vertebral and spinal cord neoplasms (63,64). For the first time, the spinal cord can be examined intrinsically directly. Intravenous injection of paramagnetic contrast medium may be helpful in MR imaging of the head and neck by revealing the perineural spread of tumor (65). MR imaging may reveal vascular lesions by demonstrating flow voids. In the spinal canal, diffusion-weighted imaging may be useful in separating benign from malignant lesions affecting the vertebral bodies.

In multiple sclerosis, although MR imaging often reveals the lesions and can aid in discriminating acute from chronic lesions, the neurologist's clinical evaluation is still the primary method of diagnosis. MR imaging, however, has proved useful in evaluating therapeutic options by means of monitoring lesion changes over time and helping to delay or prevent new lesions.

MR imaging with rapid imaging times permits functional neuroimaging, which is rapidly opening up our understanding of cerebral activity. With the availability of new imagers that have field strengths of 3 T or more and are capable of a substantial increase in the rate and quantity of data acquisition, and a two to three times increase in resolution, our understanding of cerebral activity will surely increase further (66).

The accuracy and value of MR spectroscopy should increase because of improved data acquisition, multivoxel techniques, and smaller voxels of 1 cm or less (67). By clearly showing the biochemistry of the brain, MR spectroscopy should provide information on neurologic disorders even earlier than does MR imaging, because in some cases biochemical changes occur earliest in the evolution of such diseases (68). Use of elements other than hydrogen, such as sodium, carbon, and fluorine, may be possible because of increases in the available signal with 3-T or even more powerful magnets, and this should further open new pathways of information. Already, more powerful magnets with field strengths in the range of 6–9 T are being investigated to provide MR spectroscopic information in human subjects, so biochemical changes may provide additional new and useful information (69).

Our specialty has made quantum leaps in its ability to depict the structural le-

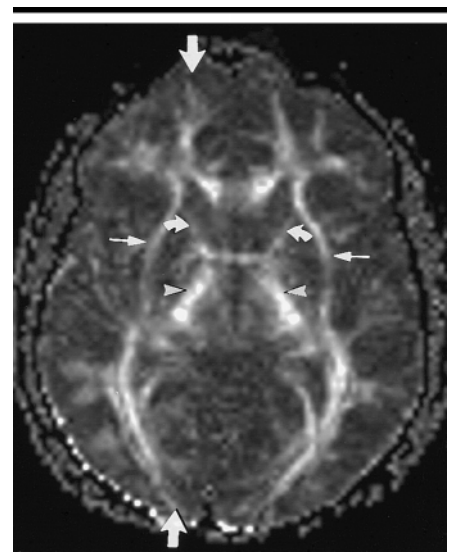


Figure 12. Transverse MR image obtained by using a diffusion tensor imaging sequence with multiple diffusion gradient orientations to take advantage of the anisotropic imaging highlighting white matter tracts. In this single image from a sequence of multiple images, the following tracts are identified: arcuate fasciculus (small straight arrows), corticospinal tract (arrowheads), internal capsule anterior limb (curved arrows), fronto-occipital tract (large straight arrows).

sions of the brain, spine, head, and neck, while reducing patient discomfort and morbidity. This progress in lesion detectability has been possible as newer modalities have provided more information. In addition, much has been learned about the natural history and evolution of disease through our ability to repeat noninvasive diagnostic examinations and to follow the patient's clinical course.

The chronology of neuroradiology, which started with the discovery of x rays by Roentgen in 1895 and his application of them in x-ray imaging, was quickly followed by early intellectual and technical advances, beginning with conventional radiography in the early 1900s, followed by ventriculography in 1918 and pneumoencephalography in 1919. The next important stride came in 1927 when angiography was developed. After this, serial changers and power injectors made improved imaging possible following direct arterial sticks. Catheter angiography, which was developed in the 1960s, replaced direct sticks.

In 1971 came the technologic advance, that for neuroradiology, was comparable with the trip to the moon by means of space flight. This was the year in which CT was introduced. A revolutionary noninvasive approach to brain and then

spine imaging was possible. We were able to look inside the brain for the first time and to observe its internal structure. Soon thereafter came MR imaging, which marked an even greater leap forward than CT. Structural imaging made substantial progress, and then came physiologic imaging and the development of functional imaging with stimulated cerebral activation, similar to positron emission tomographic scanning, as well as perfusion and diffusion.

With the advent of MR spectroscopy, we are at the beginning of biochemical imaging. The availability of newer magnets with increasing power at 3 T and then with 6–9 T will open the window of our diagnostic abilities and knowledge even wider.

We were most fortunate to start in neuroradiology during the age of angiography and the beginning of pathophysiologic studies, and then to be a part of the magical changes that have occurred since then, including CT and MR imaging. As MR imaging continues to improve, new vistas in neuroradiology will be opened up that will provide more accurate anatomic, physiologic, and biochemical information about disorders of the nervous system.

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